McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Building 155, Station Core Facility

Location ID: B155

Two-story building divided into a single-story subsistence area and a two-story dormitory area. The subsistence area houses the station dining facility, laundry, NSFA photo labs, and boiler room. The dormitory area includes dormitory rooms housing up to six persons each, television and radio studios, a store, a barber shop, and offices.



Photo 1: *Building 155, Station Core Facility*

General Structure Information



Automatic External Defibrillator (AED) Locations

This building contains one or more AED devices that may be used in the event of life-threatening cardiac arrhythmia.

- **Weight Room**
- **Highway One, across from Barber Shop**

Usage	Station core facility
Occupancy Group	A-2.1 and R-1 Assembly and Dormitories
Maximum Occupancy	1,372
Construction Type	Type V, 1-Hour, combustible; steel-framed structure with insulated metal siding and roofing, steel-framed heavy-timber first floor, steel-framed corrugated metal and plywood second floor, wood-framed gypsum board interior partitions.
Year Constructed	1969
Dimensions	165' x 320'

HVAC System

Heating	Two Bryan low-pressure steam boilers (4,800,000 Btu/hr each)

Air Handling Units	Central station air handlers with glycol heating coils. Coils in air handling units temper the air at 60 degrees F and thermostatically controlled reheat coils temper air for each room.
Heat Exchangers	Steam to glycol heat exchangers provide heated glycol for building heat.
Fuel Tank	5,500 gallon fuel tank

Water and Sewer

Water Service	4" stub off the 6" main
Sewer Service	4" line off the6" main
Connections	Sinks: 0 Toilets: 0 Urinals: 0 Showers: 14 Water Fountains: 0 18 lines with 25 instruments
Water Heater	Steam heated water tanks, two at 680 gallons each. Only one water tank is online at a time.

Electrical Service

Service Feeder	Feeder B

Electrical Service	Two 4160V, 3PH, 600A primary disconnects One 208/120V, 3PH, 750kVA transformer One 208/120V 3PH, 150kVA transformer inside building One 208/120V, 3PH, 3000A secondary main breaker One 208/120V, 3PH, 600A secondary main breaker
Phone and LAN	18 lines with 25 instruments

Fire System

Fire Panel Type	Two King-Fisher KF-DS-24 control panels, two KFRTI-WP-8 radio interface/transmitter panels, one KFRTI-20 industrial radio and control system (IRACS) panel located in boiler room. Also equipped with pull stations and alarm horns.
Heat Detectors	Attic, hallways, and non-berthing areas
Heat/Smoke Detectors	Berthing rooms
Battery-Operated Soke Detectors	Laundry room
Suppression Type	Two dry-pipe sprinkler systems, one for the berthing and store areas and the one for the kitchen and boiler room areas. Berthing: Automatic Sprinkler Company, 6-inch, Model 39, dry pipe valve with system air maintenance.

Kitchen:
Reliable Automatic Sprinkler Company, 4-inch, Model D, dry pipe valve with system air maintenance.

Both sprinkler systems release mechanically either on rate of pressure drop (through accelerator) or on low pressure (air pressure holds
clapper valve closed). Sprinkler opens on heat, causes air pressure loss, which opens clapper in dry pipe valve. 5,000 gallon water tanks,
one for each system, provides 2,500 gallons of water at 150 psi for the kitchen/boiler area and 3,300 gallons at 100 psi for the berthing/store
area. Automatic dry-chemical fire extinguishers are located over the boilers.

Floorplans

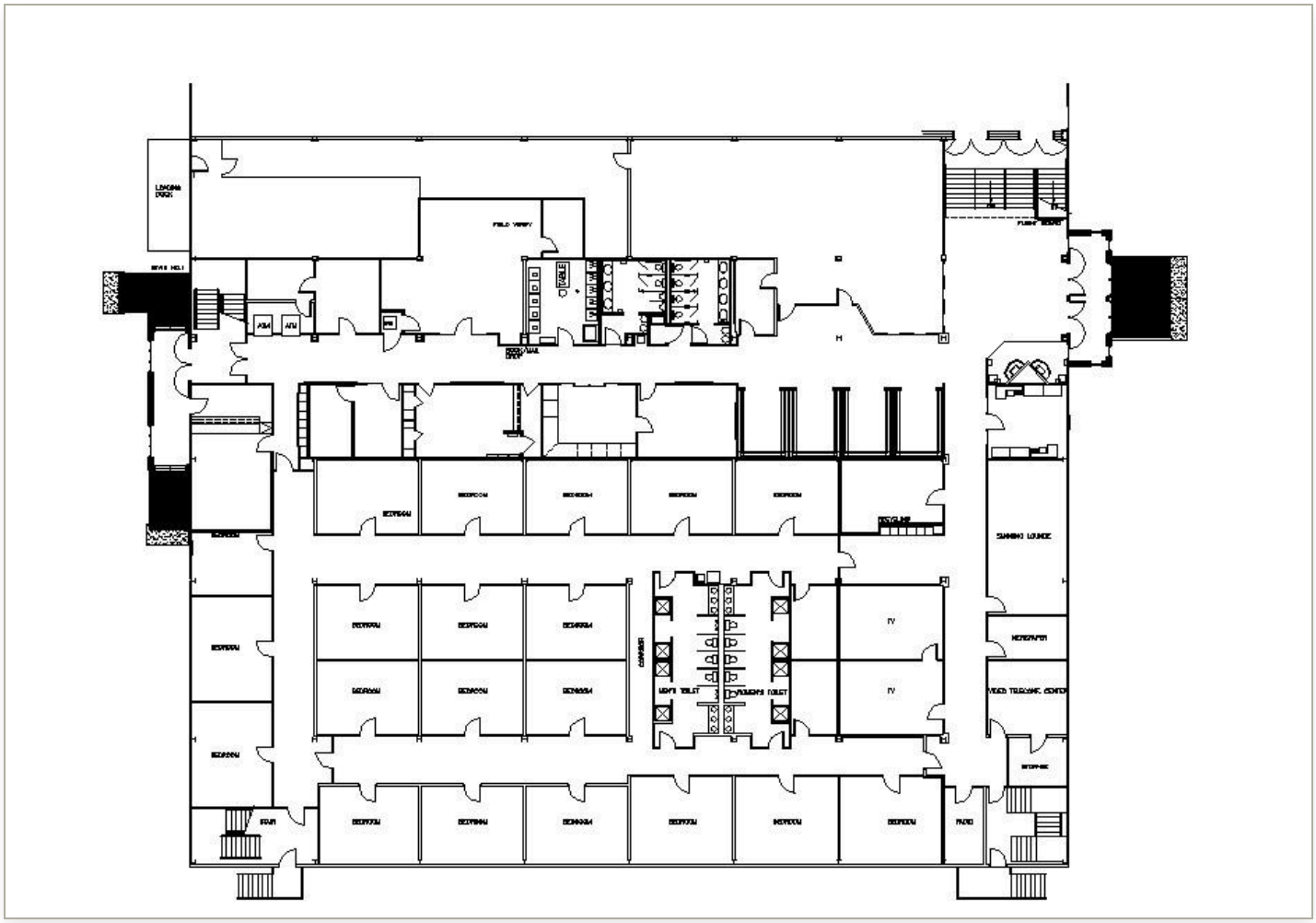


Figure 1: *Building 155, Station Core Facility*

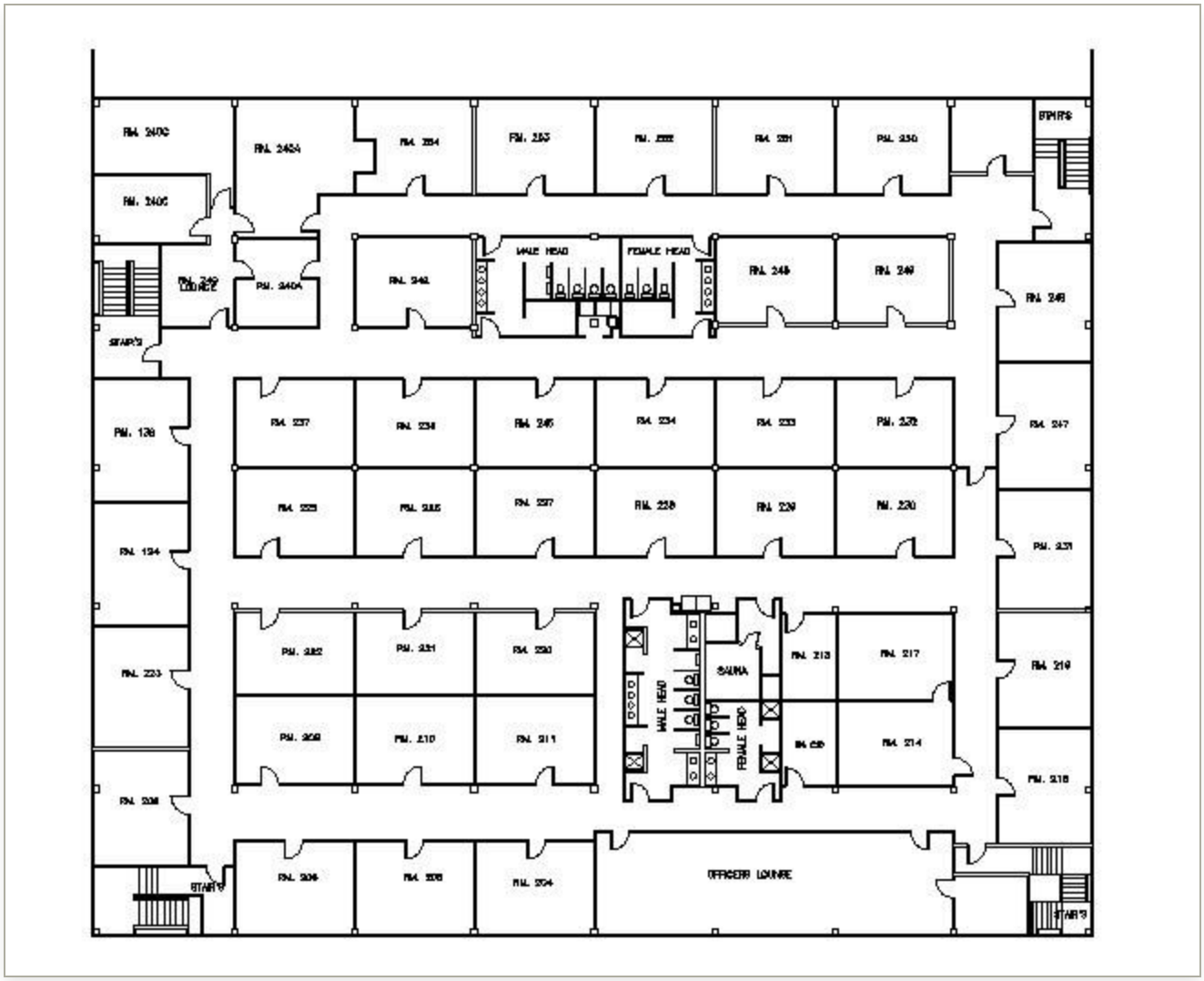


Figure 2: *Building 155, Station Core Facility*

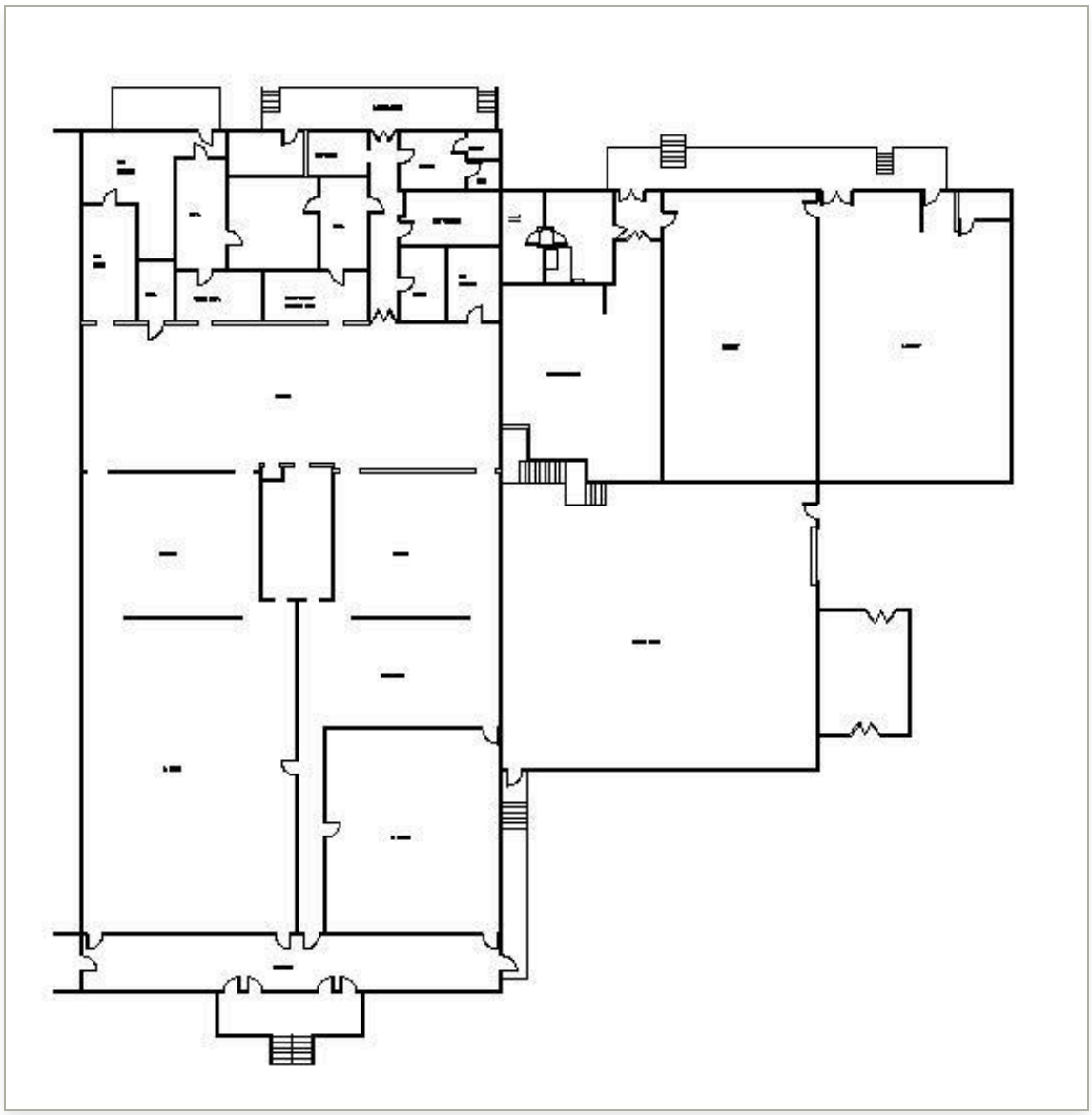


Figure 3: *Building 155, Station Core Facility*

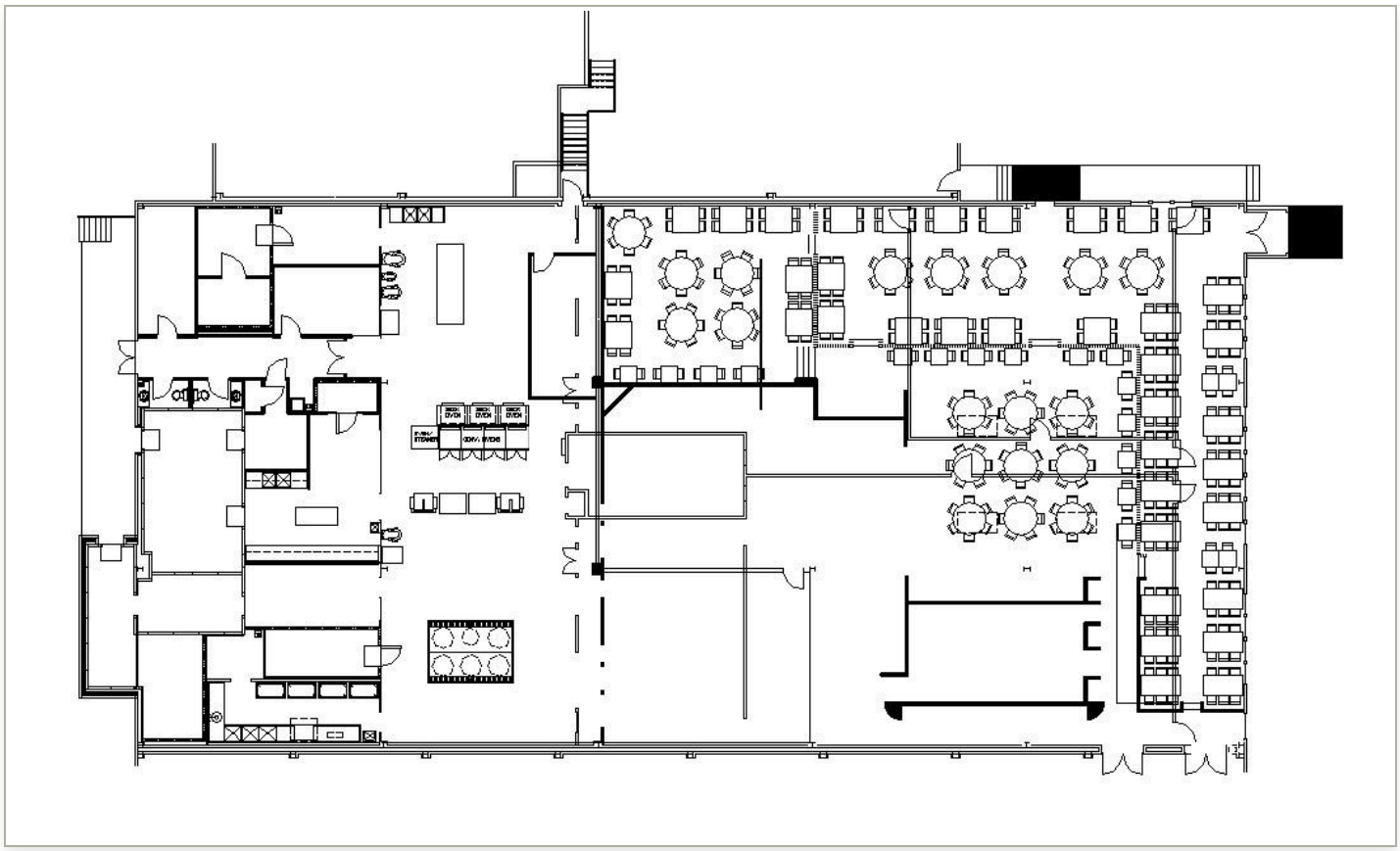


Figure 4: *Building 155, Station Core Facility*